

Stochastic actor with seed sampling

This note proposes implementation alternatives for stochastic actor policies in BAFC through the idea of seed sampling (<https://arxiv.org/abs/1802.01282>).

Overview of original paper

The original paper considers the coordination of multiple agents learning under an MDP, with shared historical observations. Here are the key concepts:

- It maintains a **posterior belief** F_t over the candidate family of MDPs at round t .
- Each agent k is associated with a **seed** $z_k \sim \nu$ which is fixed over time.
- There is a **seed-to-hypothesis mapping** G_t such that the push-forward $G_t \# \nu = F_t$.
- Each agent k acts conditioned on the **transformed seed** $G_t(z_k)$.
- Update the posterior F_{t+1} using the shared observations across agents.

Goal: Allow the multiple concurrent agents to (1) **explore** using diverse seeds, (2) be **consistent** over time via the fixed seed, and (3) **exploit** the collected data via the posterior F_t and the corresponding map G_t .

1. Across-episode exploration

1.1. Critic ensemble: view t as env_step

We do not maintain an evolving posterior over the possible worlds. Instead, we have an evolving ensemble of critics $\{Q_{t,j}\}_{j=1}^J$ trained on the collected data. The critics represent our current estimate about the values of the tuples (policy, state, action), and thereby encode our belief about the worlds. Thus we can view the (uniformly weighted) critic ensemble as our posterior belief:

$$F_t \approx \frac{1}{J} \sum_{j=1}^J Q_{t,j}.$$

1.2. Seed and actor selection

Since we do not have concurrent actors, we instead use seeds to explore across different hypotheses:

- Associate int seed j to critic j (which may seem trivial).
- At the t -th episode, draw $e_t \sim \text{Unif}(1, 2, \dots, J)$.
- Let G_t maps the critic index seed e_t to the best actor (among an ensemble) under Q_{t,e_t} (**exploit**):
 - Score each actor $S_i^{e_t} = \frac{1}{L} \sum_{\ell=1}^L Q_{t,e_t}(\pi_i, s_\ell, \pi_i(s_\ell))$.

- Select actor for rollout $i_t^* = \operatorname{argmax}_i S_i^{e_t}$.

Explore is by keep an ensemble of critics; **consistency** is by using the same actor over the episode.

2. Actor preference: view t as in-episode step

For a fixed episode n , we have the base stochastic policy π_n (or an ensemble). It is trained using the collected observations and the critic ensemble, hence representing our belief to act.

Gaussian actor: For now, consider the Gaussian actor $\pi_n(s) = (\mu_n(s), L_n(s)L_n(s)^\top)$ which gives the mean and variance. When the action space is bounded, we pass the drawn action to a squash.

We take the high-level idea from seed sampling as follows:

- Draw a seed $e_n \sim \nu_n$ at the beginning of the episode $t = 0$.
- At time $t > 0$, draw a fresh noise $\eta_t \sim N(0, 1)$.
- Set action $a_t = \mu_n(s_t) + L(s_t)(\sqrt{1 - \lambda^2}e_n + \lambda\eta_t)$.
 - When $\lambda = 1$, this is RLPD with fresh noise.
 - When $\lambda = 0$, it is fully deterministic seed-biased policy.
- The seed e_n may evolve over t within the episode, e.g. using Ornstein-Uhlenbeck process.

The idea/goal is as follows:

- **Explore** different action preferences by the seed e_n .
- **Exploit** the collected data from the learned policy π_n and the potentially evolving seed measure ν_n .
- **Consistency** is due to the fixed seed e_n over the episode and the parameter $\lambda < 1$.

2.1. Critic update

There are two possibilities when it comes to the critic update:

Deploying the base policy

This aligns with the idea that we use the seed e_n mainly for exploration purpose. Then we should (1) use the base policy π_n for critic TD updates, and (2) evaluate the actual performance using separate evaluation episodes.

Deploying the seeded policy

We will not need separate evaluation episodes for performance evaluation. In this case, we should use the seed-conditioned policy $\pi_n(\cdot; e_n)$ in the critic TD updates with the sampled seed e_n .